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Beyond the Model – The Implementation gap

Aditi Rao
Doctoral candidate, Imperial College London
a.rao24@imperial.ac.uk

AI for TB screening in India (2018 – 2020)

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- 2.4 million cases of TB in India
  540,000 cases 'missed'
- 80% of the general population & 50% of those with TB seek care in the private sector
- Target to eliminate TB by 2025
- Implementation of qXR in 2018

AI for TB screening in India (2018 – 2020)

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- Trained using > 1 million chest x-rays from India
- Area under the curve = 0.92
- Sensitivity = 90%; Specificity 70%
- Higher accuracy compared to human readers using current standard test

Source: PATH / PLOS Digital Health (2023). Implementing a chest X-ray AI tool to enhance TB screening in India: Lessons learned.

AI for TB screening in India (2018 – 2020)

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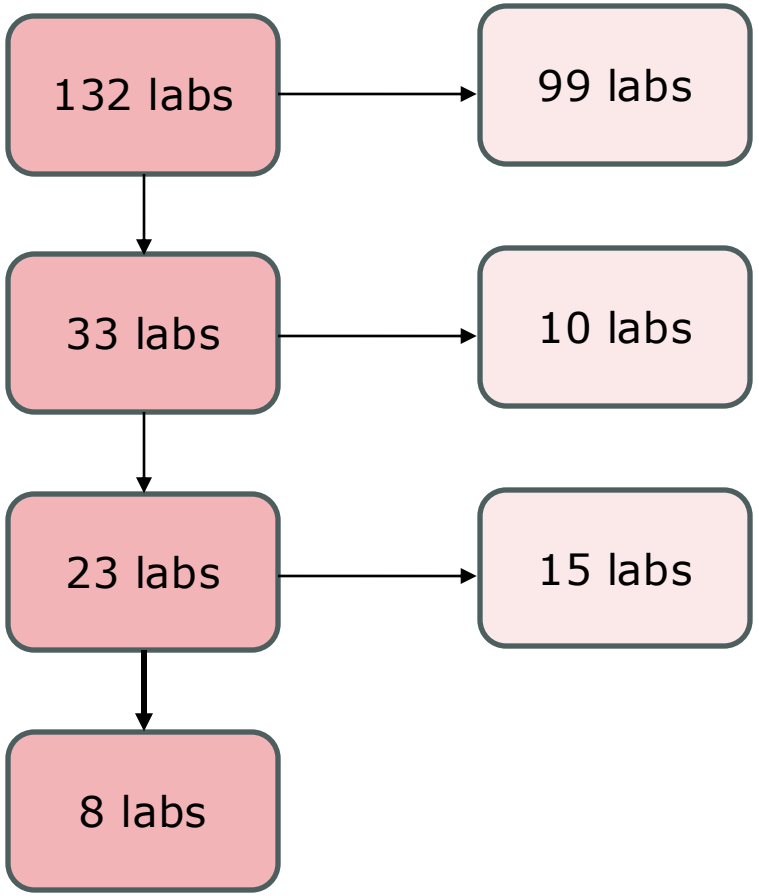

Lack of regulatory clearance → Radiology Association didn't approve installation of qXR

Fear of AI replacing doctors → Labs didn't approve installation of qXR

Lack of stable internet, local network, or static IP + transfer of administrative rights to CXRs

Source: PATH / PLOS Digital Health (2023). Implementing a chest X-ray AI tool to enhance TB screening in India: Lessons learned.

The implementation gap

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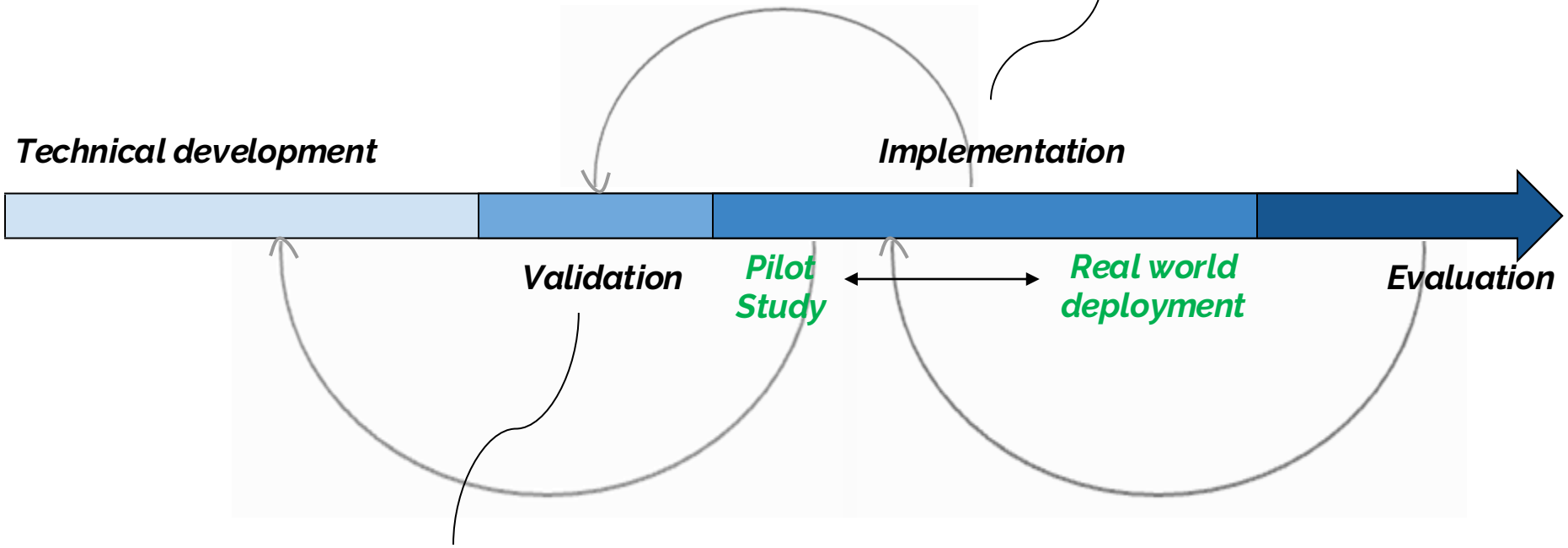
< 2% of AI-systems go beyond the development phase to implementation → **< 5%** of AI-systems that are implemented are in low-resource setting



90-95% of AI-systems developed are validated in-silico, using hypothetical or synthetic data

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EXCLUSIVE

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Google's medical AI was super accurate in a lab. Real life was a different story

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Editorial

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The Epic Sepsis Model Falls Short—The Importance of External Validation

Anand R. Habib, MD, MPhil[1,2]; Anthony L. Lin, MD[1]; Richard W. Grant, MD, MPH[3,4]

» Author Affiliations | Article Information

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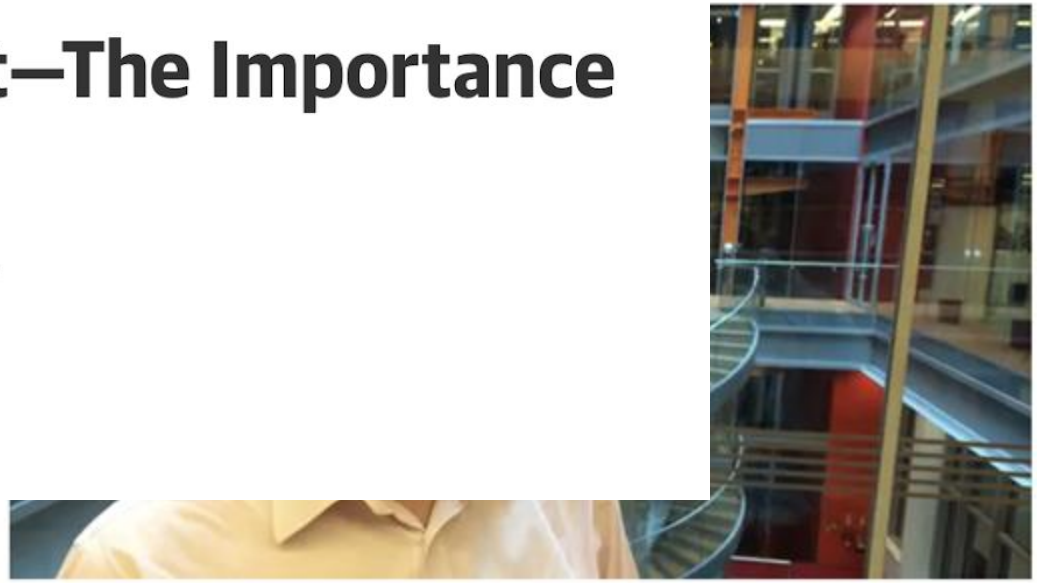

By Rory Cellan-Jones ›
Technology correspondent
@BBCRoryCJ ›

Key Challenges

1. Data

Who collected it and does it reflect your patients?

- Most published ML models are trained on data from high-income countries
- African populations represent a fraction of global training datasets – despite the highest genetic diversity (3,000 + ethnic groups)
- Algorithms may bias towards those are able to access and afford a treatment
- Missing data patterns differ – patients may arrive sicker, with fewer prior records
- Equipment variation changes x-ray quality, drug test sensitivity, waveform characteristic etc.

qXR in South Africa

Performance was strong on average — but the model performed worst on:

- ⚠️ Older patients
- ⚠️ People with HIV
- ⚠️ Previous TB cases

Exactly the patients most at risk in sub-Saharan Africa

Source: Lancet Digital Health (2024). Computer-aided detection of TB from chest radiographs in a TB prevalence survey in South Africa.

2. Infrastructure and Resources

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Connectivity

Only 31% broadband penetration in Nigeria (2018). qXR teams had to buy portable WiFi devices just to run the study. What happens post-study?

Power supply

Unreliable electricity disrupts cloud-based inference. The qXR TB stethoscope in Nigeria succeeded partly because it was battery-powered and offline-capable.

Human resources

Up to 20,000 health professionals leave Africa annually. Who maintains the model? Who retrains it? Sustainability requires local ownership, not just access.

Model complexity

Logistic regression and small ensembles (sound familiar from days 1 & 2?) often outperform deep learning when data is limited — and are far easier to audit and maintain.

3. Trust, Stakeholders and Equity

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Who needs to trust the model?

1. Clinicians
   - *Does it support my judgement or replace it?*
2. Patients
   - *Historically marginalized communities have reason to be skeptical of algorithmic tools*
3. Health system
   - *Who is accountable when the model is wrong? Regulatory pathways in LMICs are underdeveloped*
4. Funders
   - *Research teams leave. Sustainability means local ownership from day one.*

The equity problem

A model that achieves 95% aggregate accuracy can still,

⚠️ Perform at 65% for rural patients

⚠️ Systematically miss women in datasets

⚠️ Fail for older patients, HIV-positive patients etc.

4. User perception, interface design and training

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1. User perception and cultural context

- Designed in English – *not adapted to local languages and norms (localization not just translation)*
- Designs assume high digital literacy – *unintuitive*

2. Interface and structural design

- Interfaces built for keyboard and mouse – *often break on shared mobile devices or touch screens*
- Alert formats and colour coding from Western clinical contexts don't translate culturally
- Online/ 5G capability – *assumes connectivity*

3. Training and capacity building

- Training is typically led by the research team – *what happens when the grant ends*
- Training must be re-run repeatedly – *sub-Saharan Africa loses ~20,000 healthcare workers/ year*
- Designated champions – *without a local lead to support ongoing use, adoption fades within months*

Evaluation beyond technical accuracy

Levels of evaluation

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Technical

Does the model work?

- AUC, sensitivity, specificity, precision
- Calibration – are probabilities reliable?
- Subgroup sensitivity analyses – not just the aggregate
- Performance on your local population

Days 1 & 2 taught you this

Clinical & Operational

Does it change how care is delivered?

- Does the model output fit clinical workflow?
- Is the model usable and interpretable?
- Does it change clinician behaviour?
- Time-to-decision: does it truly save time?
- Alert fatigue: does it generate too many flags?

Often unanswered

Health Outcomes

Does it improve patient outcomes?

- Reduction in morbidity/ mortality
- Equity – does it improve outcomes for all groups?
- Population-level impact

Ultimate impact and why we do this

Thinking about real-world adoption

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1. Implementation frameworks
   - RE-AIM Framework: Reach, Effectiveness, Adoption, Implementation, Maintenance
   - NASSS Framework: Non adoption, Abandonment, Scale-up, Spread, and Sustainability

2. Participatory design
   - Involve the end-users – clinicians, patients, administrators – from the start
   - Co-design outputs with healthcare workers, not just data scientists
   - Test in the actual warn environment, not in a demo room

3. Efficacy $\neq$ Effectiveness
   - Most ML/AI papers published report efficacy, very few follow it to effectiveness and monitoring over time.